Review 0 Preparation Guide

An Intelligent Public Transport Delay Prediction and Route Optimization System

1. Core Narrative (Your 'Elevator Pitch')

When the panel asks 'What is this project?', focus on the novelty. Do not just say 'It predicts delay.' Explain that this is an **explainable decision support system**.

- We are solving the 'Black Box' problem in ML by using SHAP to provide natural-language explanations for delays.
- We are using a Schedule-Aware Sequential Delay Propagation (SA-SDP) framework. Delays aren't isolated; if Stop 1 is late, Stop 2 will be late. Our model captures this ripple effect.
- We provide actionable outputs via the X-MCDR routing engine, rerouting passengers dynamically and suggesting buffer times to operators.

2. Project Loopholes & Defensive Counter-Solutions

Loophole 1: Sensor Dropouts / Missing GPS Data

Defense: 'We anticipated this, which is why we built a KNN Imputation Pipeline. It uses historical averages and neighboring bus data to predict the location, keeping the system stable even with up to 15% sensor data loss.'

Loophole 2: Unpredictable Events (Accidents)

Defense: 'Correct, ML cannot predict an accident before it happens. However, our system integrates real-time traffic indices. The moment an accident causes a traffic spike, the dynamic cost model (X-MCDR) recalculates edge weights and instantly suggests a detour.'

Loophole 3: Real-time SHAP Computation Cost

Defense: 'Standard SHAP is slow. We use TreeExplainer, which is highly optimized for tree-based models like our XGBoost and LightGBM models, reducing computation time to milliseconds and allowing real-time Dashboard integration.'

3. Dataset Availability, Sources & Feasibility

1. Transit Schedules & Historical GPS Logs (GTFS & AVL Data)

Sources: Transitland (transit.land), MobilityDatabase (mobilitydatabase.org), City-Specific Portals (CTA, NYC OpenData).

2. Historical & Real-Time Weather Data

Sources: OpenWeatherMap API (openweathermap.org), NOAA.

3. Traffic Congestion Data

Sources: TomTom Traffic Stats, Kaggle Datasets.

4. Spatial Road Networks for Routing

Sources: OpenStreetMap (OSM) via OSMnx library.

4. Current Tech Stacks vs. Our Unique Architecture

To convince the panel, you must clearly contrast what existing papers and systems do versus what your system introduces. This demonstrates deep literature review comprehension.

A. Deep Neural Networks (DNN) vs. XGBoost/LightGBM + SHAP

Current Tech Stack: Many recent papers (e.g., Rum et al., 2026) use Deep Neural Networks because they offer the absolute lowest Mean Absolute Error (MAE) for delay prediction.

Our Unique Way: We actively reject pure DNNs because they are 'Black Boxes' (Gap A). An operator cannot act on a delay if they do not know what caused it. We use XGBoost and LightGBM because they provide near-DNN accuracy but allow for **TreeSHAP Explainer integration**. This uniquely allows our system to output natural-language explanations (e.g., 'Delay caused by heavy rain + traffic').

B. Isolated Predictions vs. Schedule-Aware Sequential Delay Propagation (SA-SDP)

Current Tech Stack: Standard regressors treat every transit stop prediction as an isolated event. They fail to understand that a bus delayed at Stop 1 will mathematically be delayed at Stop 2.

Our Unique Way: Instead of building extremely heavy and complex Graph Neural Networks (GNNs) to map this, we introduce **SA-SDP**. We use a tabular framework that passes the actual/predicted delay of Stop (i-1) as a live feature to predict Stop (i). This elegantly and efficiently captures the 'ripple effect' of delays down a transit line.

C. Static Timetable Routing vs. Explainable Multi-Criteria Dijkstra Routing (X-MCDR)

Current Tech Stack: Standard routing algorithms (Dijkstra/A*) used in basic applications rely on static edge weights—usually just distance or scheduled travel time.

Our Unique Way: We use an interactive OpenStreetMap graph (via OSMnx) where the edge weights are **dynamically modified in real-time** by our LightGBM regressor. Our cost formula is `Base Time + ML Predicted Delay + Weather Risk`. If a segment suddenly becomes risky, X-MCDR instantly calculates a detour and uses SHAP to explain exactly why the detour was triggered.

D. Missing Data Degradation vs. KNN Imputation Resilience

Current Tech Stack: Standard models crash or severely degrade when real-world GPS/AVL sensors drop packets (which happens 15-20% of the time in urban canyons).

Our Unique Way: We implement a **K-Nearest Neighbors (KNN) Imputer pipeline** at the ingestion layer. It uses historical segment averages and the behavior of neighboring vehicles to impute missing coordinates, allowing our ML engine to remain highly accurate even with unstable telemetry.

5. Confusing/Tricky Panel Questions & Model Answers

Q1: How is this different from Google Maps or RedBus?

Your Answer: 'Google Maps provides generic ETAs based on current traffic speeds, but it does not know transit schedules, passenger boarding delays, or schedule-aware downstream propagation. RedBus is a ticketing application, not a routing engine. Our system is specifically designed for public transit, providing *Explainable AI* (telling operators exactly why a delay is happening, like 'heavy rain + traffic') and offering

actionable dispatch recommendations.'

Q2: What is SA-SDP and why is it important?

Your Answer: 'SA-SDP stands for Schedule-Aware Sequential Delay Propagation. Most models treat transit stops as isolated events. We treat them as a chain. If a bus is delayed at Stop A, SA-SDP injects that delay as a feature when predicting the delay at Stop B. This models the real-world ripple effect of cascading delays.'

6. General Presentation Tips to Sound Confident

- **Use Technical Vocabulary:** Use terms like 'Feature Engineering', 'Geospatial Clustering', 'Downstream Propagation', and 'Multimodal Data Fusion'.
- **Focus on the Dashboard:** Remind them this isn't just a backend script. You are building a 'Dual-Perspective Dashboard' (one view for commuters, one for operators).
- **Acknowledge Scope:** If they suggest something impossible (like 'Why don't you control the traffic lights?'), say: 'That falls under intelligent infrastructure orchestration, whereas our scope is limited to predictive analytics and routing decision-support within the transit layer.'